SAQLAIN MAHMOOD

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PROFESSIONAL SUMMARY

Final-year Computer Science student with experience in hands-on AI and Generative AI, and mobile application development. Familiar with Python, React Native, TypeScript, Firebase, HuggingFace Transformers. Developed a fitness app using a RAG pipeline with LLaMA 3.3 to create personalised fitness plans using AI, and fine-tuned LLaMA 3.1 8B with QLoRA for emotion classification with 93.55% accuracy. Experienced with Git and GitHub.

EDUCATION

NUML (National University of Modern Languages) — **Sept. 2022- Jun 2026**

BS Computer Science

CGPA: 3.3

Relevant Coursework (in progress): Object-Oriented Programming, Data Structures & Algorithms, Database Systems, Software Engineering, Artificial Intelligence, Visual Programming (.Net)

TECHNICAL SKILLS

Programming Languages: C++, Python, .NET
Database Management: MySQL, SQL Server (SSMS)
Tools & Platforms: Jupyter Notebook, Visual Studio, Git, Apache, Windows, Ubuntu (basic)
Frameworks & Libraries: .NET Framework, Flutter (beginner), AI libraries (e.g., scikit-learn, pandas)

PROFESSIONAL EXPERIENCE

Systems Limited Intern — **July 2025 – August 2025**

- Developed PhotoShare, a cloud-based photo-sharing platform on Microsoft Azure, using Python with Azure Functions for serverless backend and HTML/CSS/JavaScript for the frontend.
- Integrated Azure Computer Vision for automatic AI-powered photo tagging and Azure Text Analytics for real-time sentiment analysis on user comments.
- Built scalable architecture using Cosmos DB, Blob Storage, and JWT authentication, with automated CI/CD deployment via GitHub Actions.

PROJECTS

- **Fitness Buddy** *(Final Year Project)* React Native | TypeScript | Firebase | Groq (LLaMA 3.3) | HuggingFace | RAG

- Built an AI-powered fitness app pairing users with workout partners and generating personalised workout and nutrition plans through a RAG pipeline.
- Integrated **LLaMA 3.3 70B** with HuggingFace embeddings and cosine similarity over a custom knowledge base of 73 exercises and 53 meals.
- Implemented real-time partner features including shared challenges, chat, streaks, and verified workout photos using Firebase Firestore.

- **Emotion Classification Using Fine-Tuned LLMs** (Python | PyTorch | HuggingFace | QLoRA | LLaMA 3.1)

- Fine-tuned LLaMA 3.1 8B with QLoRA (4-bit quantisation) for multi-class emotion classification, benchmarked against a TF-IDF + Logistic Regression baseline.
- Achieved 93.55% accuracy on the 6-class Emotion dataset and 57.20% on the 28-class GoEmotions dataset, outperforming traditional ML baselines by up to 16.2%.
- Addressed severe class imbalance (up to 225× ratio) and trained efficiently on Google Colab Pro Plus using Parameter-Efficient Fine-Tuning (PEFT).

- **PhotoShare Cloud-Based Photo Sharing Platform** (Python | Azure Functions | Cosmos DB | Blob Storage | Text Analytics)

- Developed a full-stack serverless photo-sharing platform on **Microsoft Azure** with role-based access for creators and consumers, deployed via **GitHub Actions CI/CD**.
- Integrated **Azure Computer Vision** for automatic AI-generated photo tags and **Azure Text Analytics** for real-time sentiment analysis on user comments (Positive / Neutral / Negative).
- Implemented **JWT authentication**, RESTful APIs with 10 endpoints, and scalable storage using **Cosmos DB** and **Blob Storage** with stateless architecture.